2022-01-21

Aerodynamics modeling

Aim: Aerodynamics data for flight simulation

Method: Nonlinear Lifting Line (NLL)

Combine detailed airfoil data with lifting line

Inviscid linear potential flow airfoil ⇒ Linear LL

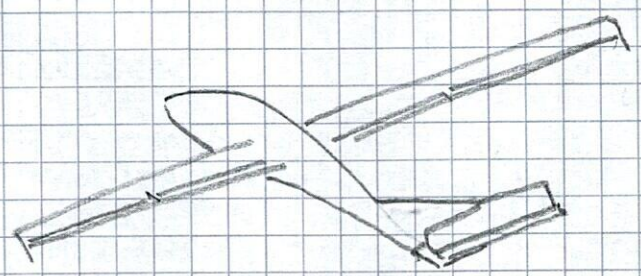
Viscid nonlinear airfoil ⇒ nonLinear LL

Limits: Low speed incompressible (Xfoil)

High aspect ratio

~ straight wing preferable

Windex



Fuselage not modeled ⇒ add corrections

Fin highly swept ⇒ Beware about lateral data

Process: 1. Generate airfoil database

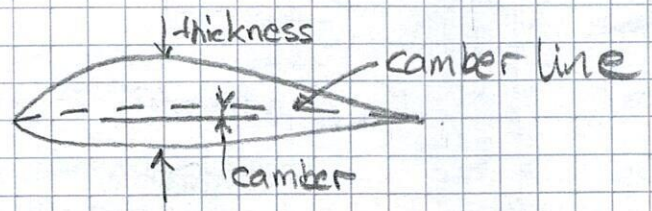
Could be simple (make_example.m simplefoil.m)

Could be experimental data

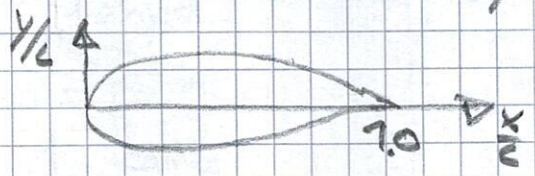
We use Xfoil viscid or inviscid

See example: make_wtm2.m buildhl5.m

Airfoil analysis



Defined in a local coordinate system



Xfoil needs sharp trailing edge

Data needs to be smooth for good results

Generate tables with:

$C_L(Re, \alpha, \delta)$

C_D

C_m

Flap setting is possible within Xfoil

Xfoil> GDES function

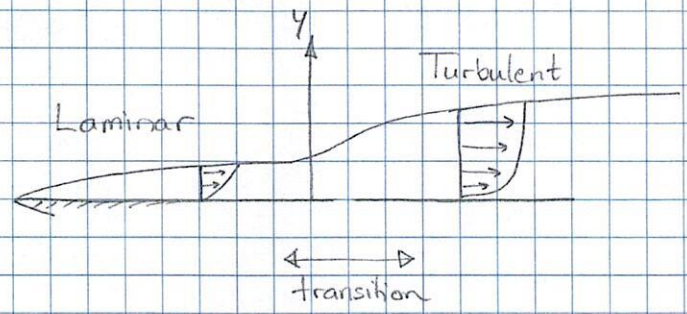
Xfoil solves for given Re, α and δ

$C_L(\alpha)$ etc by interpolation in tables
or smooth using spline interpolation
This functionality is provided
in Matlab/Octave NLL package

Read: xfoil_doc.txt user's guide
Drela's book 4.12 → 4.14

Boundary layers

Flat plate

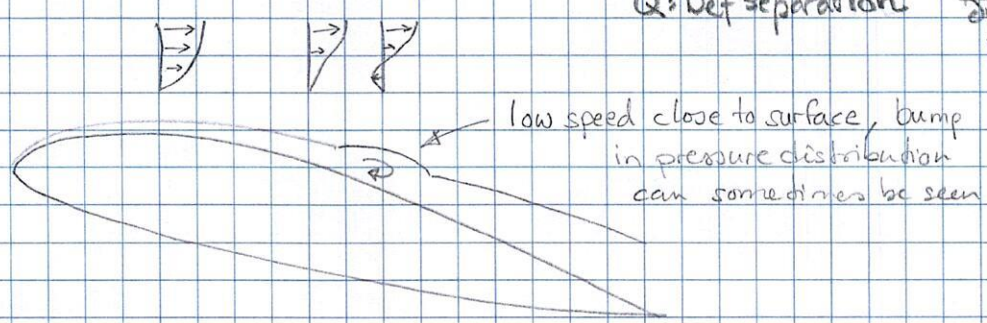


Shear stress $\tau = \mu \frac{\partial V}{\partial y}$

⇒ more shear stress in turbulent boundary layers

Q: Def separation $\left.\frac{\partial u}{\partial y}\right|_{y=0} = 0$

Separation



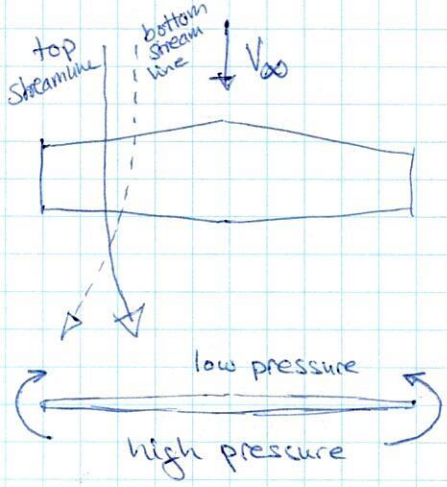
Separation bubbles cause additional drag due to the bump seen by the flow outside the boundary layer.

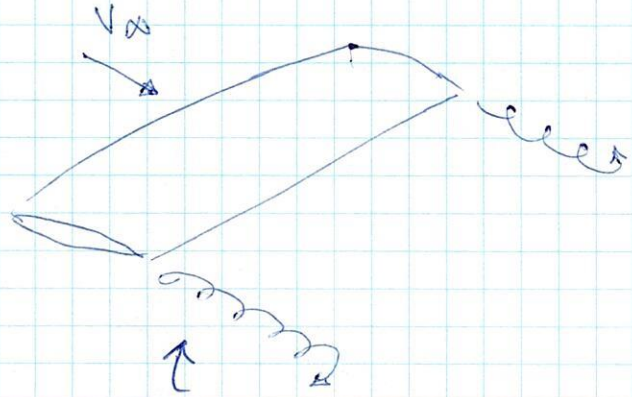
Various means for forcing transition are therefore used, in particular on gliders.

zigzag tape

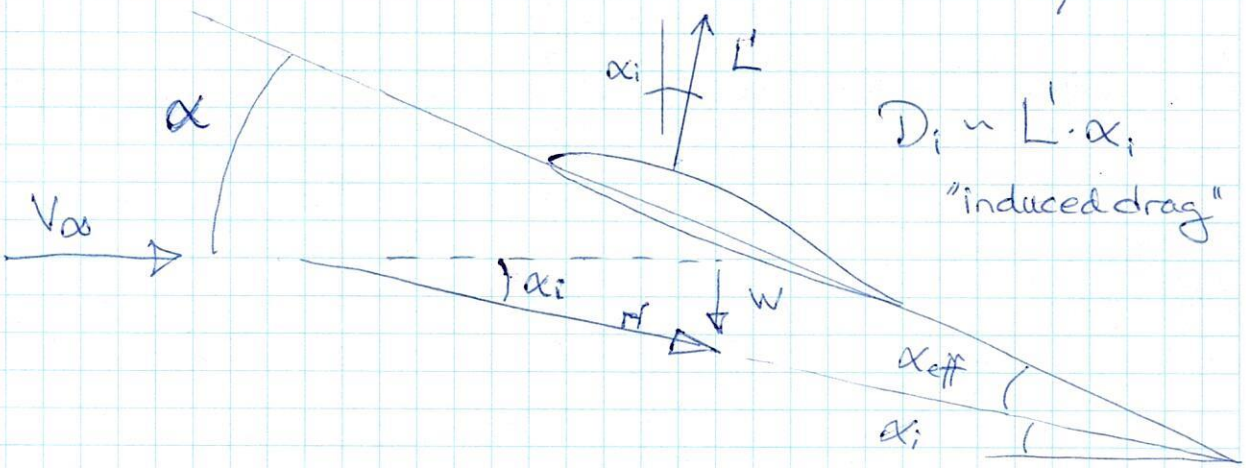
Make sure to explain: Laminar and turbulent boundary layer
Stall
Separation
Transition

Finite wing aerodynamics





Vortices create a small downward velocity "downwash"



$D_i \sim L' \cdot \alpha_i$

"induced drag"

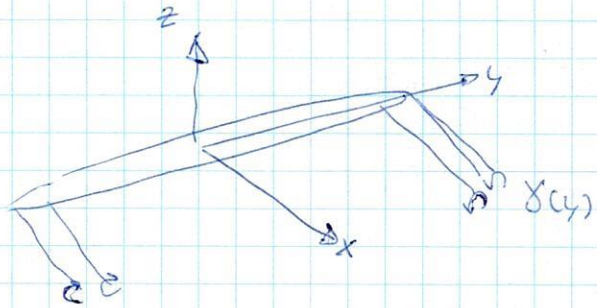
$$\alpha_{eff} = \alpha - \alpha_i$$

Lifting line Ch. 5.5 + App E

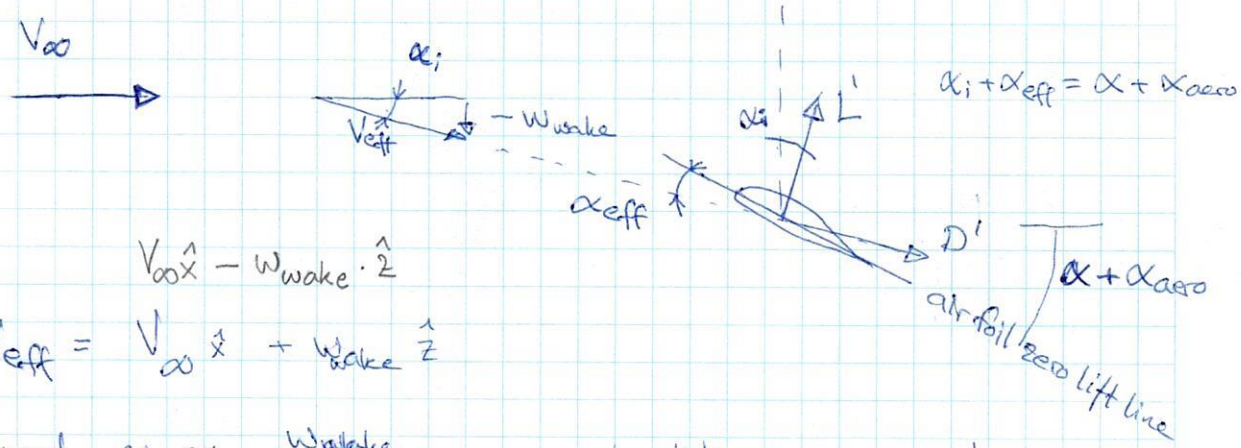
Straight wake

Wing chord distribution $c(y)$

Wing twist distribution $\alpha_{aero}(y)$



We assume sweep zero. (Provided NLL allows moderate sweep)



$\alpha_i + \alpha_{eff} = \alpha + \alpha_{aero}$

$$V_\infty \hat{x} - w_{wake} \cdot \hat{z}$$

$$V_{eff} = V_\infty \hat{x} + w_{wake} \hat{z}$$

Induced $\alpha_i \simeq -\dfrac{w_{wake}}{V_\infty}$

positive lift $\Rightarrow$ w_{wake} negative

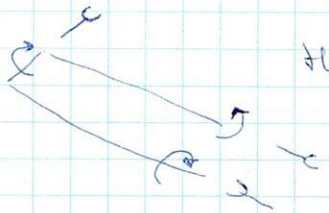
$\Rightarrow$ α_i positive

- At each spanwise station w_{wake} uniform over the chord this means α_i modifies the airfoil local angle of attack.

- At each spanwise station y, $w_{wake}(y)$ is related to the wake vortex sheet strength $\gamma(y)$

$$w_{wake}(y) = \frac{1}{4\pi} \int_{-b/2}^{b/2} \frac{\gamma(y')}{y - y'} \, dy'$$

Obtained using Biot-Savart law



Horse shoe vortex

Local 2D

force per unit length

$$L'(y) = \frac{1}{2}\rho V_\infty^2 \cdot c \cdot C_l(\alpha_{eff})$$

$$D'(y) = \frac{1}{2}\rho V_\infty^2 \cdot c \cdot C_d(\alpha_{eff})$$

from xfoil or experiment

$$\alpha_{eff}(y) = \alpha + \alpha_{aero} - \alpha_i$$

somewhat less than $\alpha + \alpha_{aero}$ due to the induced α_i

$$L = \int_{-b/2}^{b/2} (L' \cos\alpha_i - D' \sin\alpha_i)\, dy \sim \int_{-b/2}^{b/2} L'\, dy$$

$$D = \int_{-b/2}^{b/2} (D' \cos\alpha_i + L' \sin\alpha_i)\, dy \sim \int_{-b/2}^{b/2} (D' + L'\alpha_i)\, dy$$

high α
high D
$\Rightarrow$ not ok

high AR and no stall

Superimpose vortex strength $\Delta\Gamma$ to match trailing edge condition $\Rightarrow$ linear system of eqns if $C_{l\alpha}$ is constant. Otherwise a nonlinear system

or lift?

$$\Gamma = \frac{1}{2} V_\infty c \cdot c_l(\alpha + \alpha_{aero} - \alpha_i)$$

$$\alpha_i(y) = \frac{-w_{wake}}{V_\infty}$$

3st

$$2bV_\infty \sum A_n \sin(n\theta) = \frac{1}{2} V_\infty c \cdot c_{l_\alpha} \left(\alpha + \alpha_{aero} - \sum n A_n \frac{\sin(n\theta)}{\sin\theta}\right)$$

$$\sum_{n=1}^{\infty} A_n \left[\sin(n\theta) + \frac{1}{4b} c \cdot c_{l_\alpha} n \frac{\sin(n\theta)}{\sin\theta}\right] = \frac{c(\theta)}{4b} c_{l_\alpha} (\alpha + \alpha_{aero})$$

$$\int_0^{\pi} W_m(\theta)\, d\theta \qquad W_m = \delta(\theta_m) \quad \theta_m = \pi \frac{m}{N+1}$$

nonlinear:

$$\sum A_n \sin(n\theta) - \frac{c}{4b} c_l \left(\alpha + \alpha_{aero} - \sum n A_n \frac{\sin(n\theta)}{\sin\theta}\right) = 0$$

NLL = Solve a system of <u>nonlinear</u> equations for A_n

c_l smooth of α helps...

KUNGL TEKNISKA HÖGSKOLAN

Blad nr ..... Uppgift nr .....

Namn Estimate C_{lp} Person nr 2019-02-05

$$\hat{p} = \frac{pb}{2u} \qquad C_l = C_{lp} \cdot \hat{p}$$

$$\alpha = \frac{w}{u} \qquad w = p \cdot y$$

$$\alpha = \frac{py}{u} = \frac{2u}{b} \cdot \hat{p} \cdot \frac{y}{u} = \frac{2y}{b} \cdot \hat{p}$$

section lift

$$\sin 2x = 2 \sin x \cos x$$

$$y = \frac{b}{2} \cos\theta$$

$$dy = -\frac{b}{2} \sin\theta \, d\theta$$

trapz (x,y)

$$L = \rho V_\infty \int_0^\pi 2b V_\infty \sum_{n=1}^{\infty} A_n \sin(n\theta) \frac{b}{2} \sin\theta \, d\theta$$

$$L = \rho V_\infty \int_0^\pi 2b V_\infty \sum_{n=1}^{\infty} A_n \sin(n\theta) \cdot \underbrace{\frac{b}{2}\cos\theta}_{y} \cdot \frac{b}{2} \sin\theta \, d\theta$$

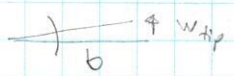
$$= \rho V_\infty^2 b^3 \frac{1}{2} \int_0^\pi \sum A_n \sin(n\theta) \cdot \underbrace{\cos\theta \cdot \sin\theta}_{\frac{1}{2}\sin 2\theta} \, d\theta$$

$$\int_0^\pi A_2 \sin 2\theta \cdot \frac{1}{2} \sin 2\theta \, d\theta = \frac{\pi}{4} \cdot A_2$$

$$L = \frac{1}{2} \rho V_\infty^2 \cdot b^3 \cdot \frac{\pi}{4} \cdot A_2$$

$$C_l = \frac{L}{\frac{1}{2}\rho V_\infty^2 \cdot S \cdot b} = \frac{b^2}{S} \cdot \frac{\pi}{4} A_2 = AR \cdot \frac{\pi}{4} \cdot A_2$$

Wt-test freq = 3 Hz $\qquad \hat{p} = \frac{b}{2u} \cdot p$



$$w_{tip} = b\dot{\phi}$$

$$\phi = \hat{\phi} \sin \omega t$$

$$\dot{\phi} = \hat{\phi} \omega \cos \omega t$$

$$\alpha_{tip} = \frac{w}{u} = \frac{b\hat{\phi}\omega}{u}$$

$$\frac{1.1}{20} \cdot \frac{10}{180} \cdot \pi \cdot 3 \cdot 2\pi \approx 9^\circ$$

Plan
teari
case 500 mm
simple LL
xflr5
NLL

Lifting line modeling

Read: Drela's book Appendix E and 5.5
simple model in liftline.m in folder finite_wing

Model the wing and compare with wind tunnel test data

Suggestion:

Modify nonlinear_lifting_line/
main_simple.m
make_wtm2.m

Geometry: finite_wing/wtmodel_planform.pdf

Compare: finite_wing/Matlab_tunneldata

Remember: The wind tunnel model uses $\delta = -5^\circ$

After fs = flight_state(...)
set: fs.delta = fs.delta - 5 pi/180

Plot: C_L and C_m vs α
C_L vs C_D